FOODBRIDGE

Connecting surplus food with the people who need it

Team Project Proposal | Submitted on 3 November 2026

DETAIL	INFORMATION
Course / Subject	B. Tech
Submitted to	Mr. Vivek
Team members	Piyush (Team Lead),Shikhar, Vansh, Vikas , Yudhishthir

1. Our idea in simple words

Every day, restaurants, college canteens, hostels and event halls are left with good food that nobody will eat. At the same time, NGOs are looking for food for the people they care for. The food often goes to waste simply because the two sides cannot find each other in time.

FoodBridge is a website that fixes this. A person with extra food posts it on the site. Nearby NGOs can see it, one of them accepts it, picks it up, and confirms that it was delivered. Everything happens in one place, and the website keeps a record.

2. Why we chose this project

- **Food is wasted while people go hungry.** Cooked food stays safe for only a few hours, so speed matters.
- **Today it depends on phone calls and messages.** It is hard to know who is free, who is close, and whether the food really reached someone.
- **It is a real problem we can actually build.** It lets our team practise full-stack web development, databases, user roles and testing, and it does some good too.
- **It fits the rules in India.** FSSAI (the food safety authority) has surplus-food regulations that encourage donation and ask for proper records. We will keep our records in the same spirit.

3. Who will use it

USER	WHAT THEY WILL DO
Food donor	Posts the food: what it is, how much, when it was cooked, until when it is safe, and where to collect it.
NGO	Registers, gets verified, sees nearby food, accepts what it can handle, books a pickup time and confirms delivery.
Admin	Checks that donors and NGOs are genuine, and handles any complaints.

4. How it will work

Donor posts food → site shows the best nearby NGOs → one NGO accepts and books a pickup → food is collected → delivery is confirmed → record is saved.

A few rules keep it safe and fair:

- Only one NGO can accept a donation. Once someone accepts, others see that it is taken.
- Every donation has a safe-use deadline. When time is up, it disappears automatically and cannot be accepted. Accepting never extends the deadline.
- If an NGO accepts but does not collect in time, the food is released so another NGO can take it.
- Exact addresses and phone numbers are shown only after an NGO accepts.

5. Main features

FEATURE	WHAT IT DOES
Smart suggestions	Ranks NGOs by distance, how much food they can take, and how much time is left. This uses clear rules, not AI, so we can explain every result.
Countdown timer	Shows how long the food stays safe, so everyone knows how urgent it is.

Edit with WPS Office

FEATURE	WHAT IT DOES
Pickup booking	The NGO picks a time slot that suits the donor.
Status tracking	Everyone can see where a donation is: posted, accepted, collected or delivered.
Admin verification	Only checked NGOs can accept food.
Impact summary	Shows how many donations were completed and how much food was delivered, using real saved records.

Extras if time allows: map view, email reminders, QR code at handover, volunteer delivery role.

6. Tools we will use

We will use the MERN stack, because all of it is written in JavaScript and our whole team can work together easily. On top of it, we will add Redis and BullMQ to handle timers and background work reliably. Section 7 explains why we chose each tool.

PART	TOOL
Website (what users see)	React.js, HTML, CSS
Server (the rules and logic)	Node.js and Express.js
Database (where information is stored)	MongoDB
Fast memory store	Redis
Background jobs and timers	BullMQ (runs on top of Redis)
Login and security	JWT tokens, with passwords stored in encrypted (hashed) form
Maps (optional)	Leaflet and OpenStreetMap, both free
Teamwork and hosting	GitHub for code sharing; a free hosting service for the demo

7. Why we chose these technologies

Each tool below is there for a reason connected to a real need in FoodBridge, not just because it is popular. Redis and BullMQ are the two newest additions, and they exist mainly to make our time-based rules (expiry and release) and our "only one NGO can accept" rule work reliably.

TOOL	WHY WE USE IT
React.js	Pages update instantly without reloading, which suits live things like the countdown timer and donation status. It is also split into small components, so each team member can build their own screens without clashing.
Node.js and Express.js	The server is in the same language (JavaScript) as the website, so everyone in the team can read and fix both sides. Express keeps the rules simple: who can post, who can accept, and the role checks.
MongoDB	Donations, NGOs and users are stored as flexible records, so we can add details like allergens or veg/non-veg without redesigning tables. It also supports location searches, which we need to find nearby NGOs.
JWT and hashed passwords	JWT lets the server know who is logged in and which role they have (donor, NGO or admin) without extra lookups on every click. Hashing passwords means even we cannot read them, and a database leak would not expose them.
Redis	Redis is a very fast store that keeps data in memory. We use it for two things. First, as a lock when an NGO clicks "Accept", so that if two NGOs click at the same moment only one wins and the other sees that the food is taken. Second, to keep frequently viewed lists (like nearby open donations) ready, so the database is not hit again and again. BullMQ also needs Redis to store its jobs.

Edit with WPS Office

TOOL	WHY WE USE IT
BullMQ	Our rules depend on time: food must expire at its safe-use deadline, and an accepted donation must be released if the NGO does not collect it in time. BullMQ lets us schedule a delayed job for each of these events, so they happen on time even if nobody opens the website. It retries failed jobs and survives a server restart, and the same system can send the email reminders listed in our extras without slowing down the main website.
Leaflet and OpenStreetMap	Both are free and need no payment details or API billing, which suits a student project. They let us show donations and NGOs on a map.
GitHub and free hosting	GitHub lets five people work on the same code with a history of changes and reviews. Free hosting lets us demonstrate a live, working website at no cost.

8. Food safety and privacy

- Donors must enter the cooking time, the safe-use deadline and how the food was stored, and confirm it was handled hygienically.
- Allergen and veg/non-veg details are shown before an NGO accepts.
- Our website only helps people coordinate. It cannot guarantee that food is safe, so people must still check the food and follow FSSAI guidance.
- Passwords are encrypted, and each type of user can only do what their role allows.

9. Who does what

TEAM MEMBER	MAIN WORK
Shikhar	Design of the pages, joining everyone's work, GitHub, final presentation
Yudhishthir	Server, login, database
Piyush	Donor and NGO pages, accepting and status screens
Vikas	Matching, pickup slots, countdown and expiry
Vansh	Admin panel, impact summary, testing, project report

Everyone will also test and review at least one other person's work, so we all understand the whole project.

10. Our plan (four weeks after approval)

WEEK	WHAT WE WILL DO
Week 1	Finalise features, draw the page designs, plan the database, set up GitHub, build the login.
Week 2	Build the main flow: post food, see listings, accept, book pickup.
Week 3	Add suggestions, countdown and expiry, admin verification and the impact summary.
Week 4	Testing, fixing bugs, putting it online, preparing the report and presentation.

11. How we will show it works

In our demo we will post a donation of 80 meals, show the suggested NGOs, accept it and book a pickup, try to accept it with a second NGO (which will be blocked), confirm delivery, and show the summary updating. We will also show that expired food cannot be accepted. We will test wrong inputs, cancellations and phone-sized screens too.

Expected result: a working, mobile-friendly website that lists, matches, tracks and records surplus food donations from start to finish.

12. Where we would like your guidance

- Is the size of the project right for our team and time?

Edit with WPS Office

- Should we keep the volunteer role and map view, or focus only on the main flow?
- Is there anything more you would like to see in the documentation or demo?

References: fssai.gov.in | react.dev | nodejs.org | expressjs.com | mongodb.com/docs | redis.io/docs | docs.bullmq.io | leafletjs.com | openstreetmap.org

Edit with WPS Office